جامعة التقنية
والعلوم التطبيقية
University of Technology
and Applied Sciences

Title:

Store Sales Forecasting Dataset

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Abstract

This study examines sales forecasting through an analysis of a dataset comprising sales transactions from various retail stores. The dataset encompasses critical information, including order dates, shipping methodologies, customer demographics, product categories, and financial metrics such as sales figures. By investigating numerical features, including sales, quantity, discount, and profit, this research aims to identify the fundamental factors that influence sales performance. The application of statistical methods for feature selection will facilitate the identification of the most significant predictors, ultimately enhancing the accuracy of sales forecasts.

Introduction

Sales forecasting represents a vital component of retail management, as it enables organizations to anticipate future sales and make informed strategic decisions. This study employs a comprehensive dataset that captures essential details regarding sales transactions across multiple stores. By analyzing the interplay between various factors and sales performance, this research seeks to uncover valuable insights that can inform effective decision-making. The analysis utilizes advanced statistical techniques for feature selection to determine the most influential variables affecting sales, thereby improving forecasting accuracy and supporting strategic planning within retail operations. This dataset utilized in this study comprises 2,121 rows and 21 columns, encompassing sales transactions from various retail establishments over an extended period. It spans from January 1, 2014, to December 31, 2017, thereby providing a comprehensive overview of sales activities throughout these years. Each data entry includes critical information, such as order dates, shipping dates, customer details, product categories, and sales performance metrics. This extensive dataset facilitates an in-depth analysis of sales trends and customer behavior, thereby supporting effective sales forecasting and informed strategic decision-making for retailers.

Purpose

The objective of this study is to develop an effective sales forecasting model utilizing a sales transaction dataset. Through the analysis of historical sales data, this research aims to identify critical factors that influence sales performance, including customer demographics, product categories, and pricing strategies. The insights derived from this analysis will assist businesses in making informed decisions regarding inventory management, marketing strategies, and resource allocation, ultimately leading to enhanced profitability and operational efficiency. Furthermore,

this study aspires to improve the accuracy of sales forecasts, thereby enabling retailers to more effectively meet customer demand and optimize their sales processes.

Analysis

In this project, I performed an analysis of sales data to uncover key factors that influence sales performance. The analysis involved several key steps:

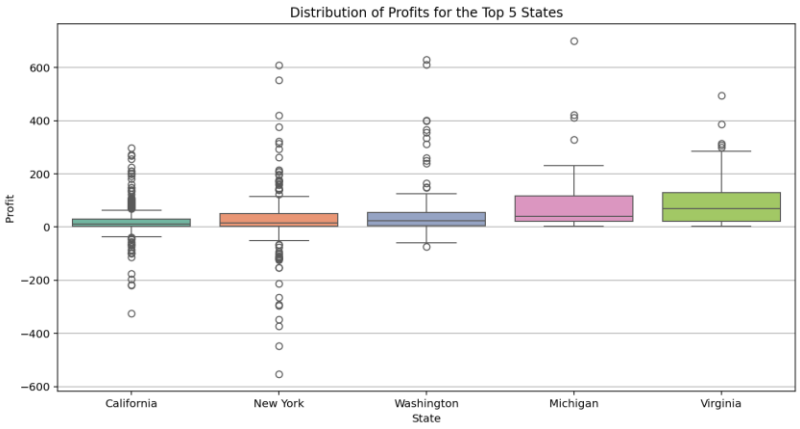
Exploratory Data Analysis:

1. **What is the impact of discount levels on total sales and profit for different shipping methods?**



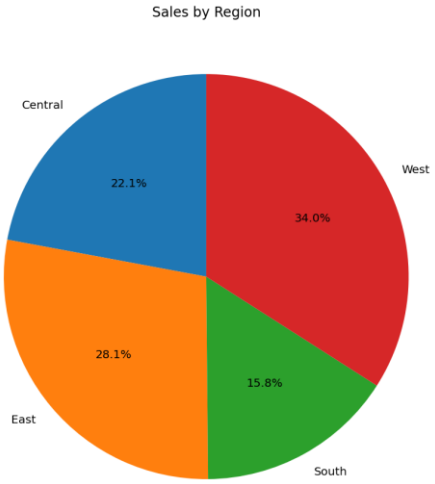
- **Figure 1: line Plot of Total Sales and Profit by Discount Levels**
    - **Explanation**: The line plot presented the impact of different discount levels on total sales and total profit. The Line plot you created displays the average profit for each shipping mode. This information is to make decisions about which shipping methods to promote or use more frequently.

2. **Distribution of Profits for Top 5 States**



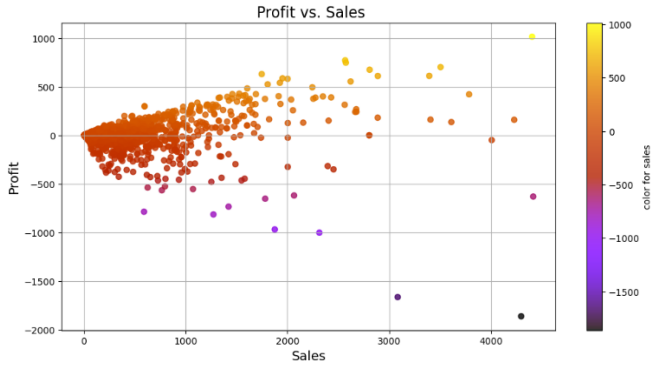
- **Figure 2: Box Plot of Profits by State**
    - **Explanation**: The box plot will show the distribution of profits for each of the top 5 states, allowing you to compare medians, interquartile ranges, and identify any outliers.

3. **Which region has the most sales?**



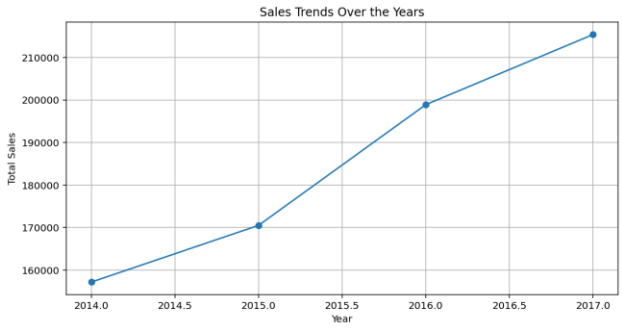
- **Figure 3: Pie Chart for Which of Sales by Region**
    - **Explanation**: This bar chart illustrates which region has the most sales, providing insights into geographic performance and potential market focus areas.

4. **What is the relationship between sales and profit, and how can it inform pricing strategies?**



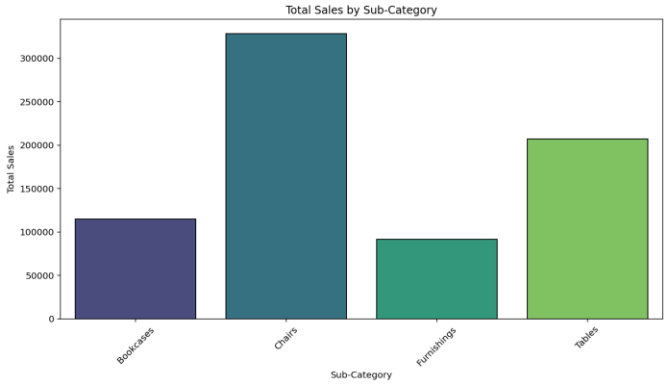
- **Figure 4: Scatter Plot of Sales vs. Profit**
    - **Explanation**: The scatter plot illustrates the relationship between sales and profit. As we note, the higher the sales, the lower the profit. It can help the business understand the pricing sensitivity of their products or services. This can lead to improved financial performance. As you see weak relationship between them.

5. **What are the sales trends over the past years?**



- **Figure 5: Line Chart of Sales Trends Over Time**
    - **Explanation**: This line chart captures sales trends over the past years, highlighting seasonal patterns and overall growth trajectories.

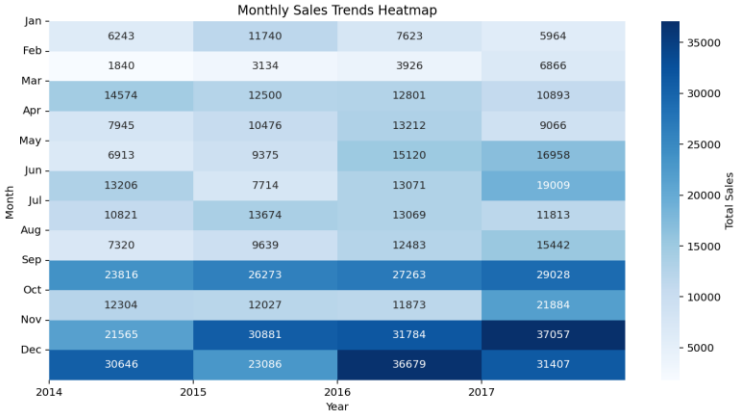
6. **Which sub-categories are driving growth?**



- **Figure 6: Bar Chart of Sales by Product Category**

- **Explanation**: This bar chart identifies which sub-categories are driving growth, focusing on top performers.

7. **How do seasonal patterns affect sales?**



- **Figure 7: Seasonal Sales Trends heatmap Chart**

- **Explanation**: This heatmap chart visualizes how months affect sales, showing peaks and troughs in sales performance throughout four years. As we can see from September to December from each year, the sales increased.

8. **How does the quantity sell correlate with the total sales revenue? Does selling more units typically result in higher sales?**

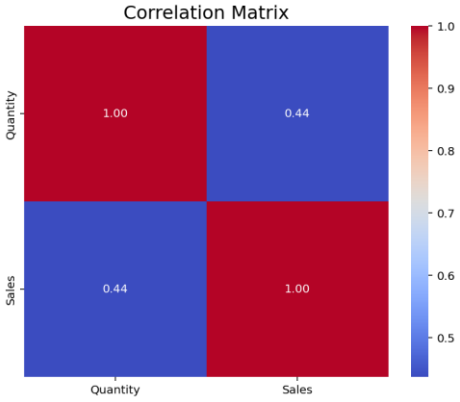
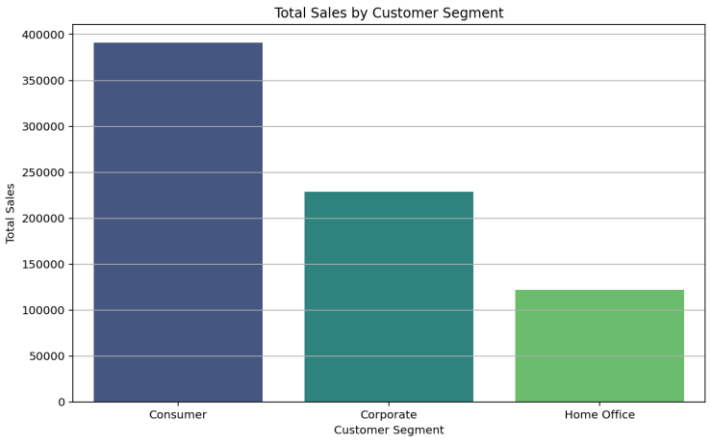


Figure 8: Scatter Plot of Quantity Sold vs. Total Sales Revenue

- **Explanation**: The correlation matrix illustrates the relationship between Quantity and Sales. With a correlation coefficient of 0.44, there is a moderate positive relationship, suggesting that selling more units generally results in higher sales, though it's essential to consider other influencing factors for a comprehensive analysis.

9. **How do sales figures correlate with different customer segments (e.g., Consumer, Corporate, Home Office)?**



- **Figure 9: Bar Chart of Sales by Customer Segment**
    - **Explanation**: This bar chart shows how sales figures correlate with different customer segments (e.g., Consumer, Corporate, Home Office), providing insights into customer behavior.

Summary

In this section, I have addressed key questions regarding the sales data through various plots and visualizations. Each question is accompanied by a figure that illustrates the findings, along with explanations that clarify the significance of the results. This structured approach allows for a comprehensive understanding of the data and its implications for business strategies.

Feature Selection:

In this process, I identified and selected relevant features that exert a significant influence on sales amounts. Key variables such as quantity sold and unit price were chosen due to their direct correlation with revenue, while less informative features were excluded. Feature selection involves the systematic identification of the most pertinent variables that contribute to the target variable, in this instance, sales. I employed statistical methods, including correlation analysis and linear regression techniques available in scikit-learn, to evaluate the relationship between each feature and the target variable. The use of Python's pandas for data manipulation, alongside scikit-learn for implementing feature selection methodologies, is standard practice in this context.

Regression Modeling:

I constructed a regression model aimed at predicting total sales based on the identified features. This model serves to quantitatively articulate the relationships between sales figures and the

relevant influencing factors. I selected Linear Regression as the appropriate algorithm to effectively fit the model to the training data utilizing the chosen features. This phase involved the use of various libraries, including scikit-learn for model development and evaluation, as well as statsmodels for comprehensive statistical analysis.

Model Evaluation:

Subsequently, I assessed the performance of the model through the application of statistical measures to confirm its accuracy and reliability in making predictions. The primary objective of regression modeling is to create a mathematical framework that relates the selected features to the target variable, thereby facilitating future sales predictions. Model evaluation is critical in verifying the effectiveness of the regression model in predicting the target variable accurately. To this end, I utilized metrics such as Mean Absolute Error (MAE), Mean Squared Error (MSE), and R-squared to quantify the model's accuracy and dependability. Additionally, tools provided by scikit-learn enabled the computation of these metrics and facilitated the visual assessment of the model's performance.

The dataset includes important features:

- **Quantity Sold**: The number of items sold in each transaction, which directly affects total sales.
- **Shipping Mode**: The delivery method chosen by customers, which can influence their satisfaction and purchasing choices.
- **Sales**: This is the target variable we aim to predict. Including it is essential for training our model to understand what influences sales amounts.

Excluded Features:

- **Order Date:** While it may show trends, it doesn't directly predict sales. It could be transformed into more useful features (like month).
- **Customer ID:** This categorical feature doesn’t provide useful insights into sales trends and could lead to overfitting.
- **Product Category:** Adds complexity without clear relationships to sales and is better analyzed separately if needed.
- **Discounts:** If inconsistently recorded, it could introduce noise and reduce model accuracy.

Our goal is to use regression analysis to understand how these features relate to total sales. This will help us build a model that can predict sales amounts based on the selected factors.

Conclusion:

This analysis aims to enhance the accuracy of sales forecasting by identifying the principal factors that influence sales performance. Through the application of feature selection techniques, the objective is to isolate the most pertinent variables within the dataset, which encompasses customer demographics, product categories, and sales metrics. This focused approach will facilitate more effective regression modeling, thereby allowing for improved predictions of future sales based on historical data. Ultimately, the goal is to deliver actionable insights that can assist businesses in optimizing their strategies and enhancing overall profitability.